Team Vision_Report

PS-1: OPENROCKET EXTRATERRESTRIAL FLIGHT DESIGN CHALLENGE

Operation Ember Ascent: Launch from Mustafar

Team Name: Team Vision

Team Members : Abhijeet, Priyanshu, Kanishk, Aditya

Date: 25/10/2026

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1. Introduction

Mustafar is a fictional terrestrial planet with an atmosphere suitable for short-duration rocket flight. The Agency requires a two-stage sounding rocket that reaches an apogee of approximately 4 km and deploys a 1 kg scientific payload at that altitude (4000 m). The rocket must launch vertically, reach apogee, deploy the payload, and then recover both the payload and the rocket safely under parachute.

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2. Mission Requirements

Target Requirements

Parameter	Required Value	Achieved Value
Apogee	3900-4100 m	3999m
Payload Mass	0.95-1.05 kg	1.0kg
Rocket Length	≤ 2200 mm	2110mm
Total Impulse	≤ 2800 N·s	2490

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3. Mustafar Environment Setup

Environmental Parameters Entered in OpenRocket

Parameter	Mustafar Value
Surface Gravity	6.50 m/s^2
Surface Pressure	75 kPa
Surface Temperature	700 K

Wind Velocity	10 m/s
Wind Direction	270°
Launch Altitude	0 m

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4. Rocket Design

Vehicle Configuration

Stage 1 Details

- Motor: AeroTech J1265T
- Diameter: 54mm
- Length:397mm
- Mass: 1095g

Stage 2 Details

- Motor: AeroTech K185W 1379
- Diameter: 54mm
- Length:437mm
- Mass:1434g

Payload Section

- Payload Mass: 1.0Kg
- Payload Type: Scientific Payload
- Deployment Event: Apogee

Recovery System

- Drogue Parachute Diameter: 45cm
- Main Parachute Diameter: 220cm
- Deployment Condition: Apogee

Body Material

Selected Material:

Cardboard

Total Vehicle Mass: 7073g

Maximum Allowed Mass (based on material): 10kg

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5. Design Approach

Rocket Geometry Selection

During making the rocket I use nose cone,body tube insuring to not exceed required length and weight of the Rocket.

Motor Selection

I used two motors by taking care of it not exceed the impulse and reach the apogee of 4000m.

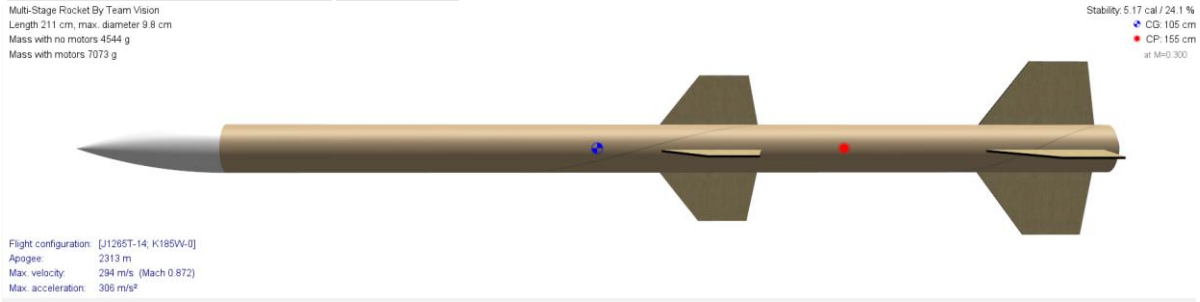
Stability Considerations

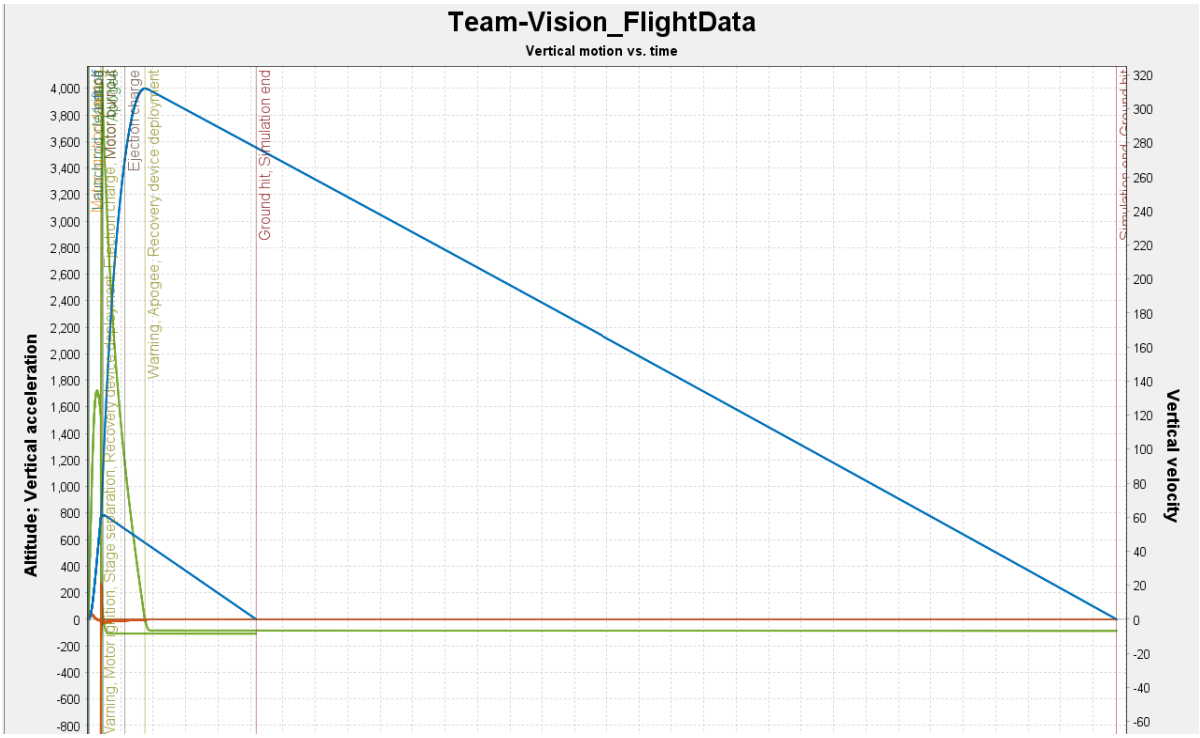
The stability of the Rocket is 5.17 cal / 24.1 %. Having CG of 105cm and CP of 155cm.

6. Simulation Results

Flight Performance

Parameter	Result
Maximum Altitude (Apogee)	3999m
Maximum Velocity	357m/s
Maximum Acceleration	310m/s^2
Flight Time	634s





7. Earth vs Mustafar Comparison

Earth Environment

Parameter	Earth
Gravity	9.81 m/s^2
Pressure	101.3 kPa
Temperature	288 K (15°C)
Density	1.225 kg/m^3

Comparison Results

Parameter	Earth Simulation	Mustafar Simulation
Apogee	2313m	3999m
Maximum Velocity	294m/s	357m/s
Stability Margin	1.65 cal	1.72 cal
Drift Distance	185m	340m
Descent Rate	5.8m/s	7.1m/s

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8. Environmental Analysis

Effect of Lower Gravity

Hypothesis

Mustafar's gravity (6.5 m/s^2) is lower than Earth's gravity (9.81 m/s^2). Therefore, the rocket should experience less weight, allowing it to accelerate more efficiently and reach a greater altitude.

Observation

The rocket reached an apogee of **2313 m** under Earth conditions and **3999 m** under Mustafar conditions.

Effect of Lower Atmospheric Density

Hypothesis

Mustafar's atmospheric density (0.374 kg/m^3) is much lower than Earth's atmospheric density (1.225 kg/m^3). Reduced density should decrease aerodynamic drag and increase altitude.

Observation

The maximum velocity increased from **294 m/s** on Earth to **357 m/s** on Mustafar. The rocket also achieved a much higher apogee.

Effect of Lower Pressure

Hypothesis

Lower atmospheric pressure reduces air resistance and aerodynamic forces acting on the rocket.

Observation

The Mustafar simulation produced higher velocity and altitude compared to the Earth simulation.

Effect of Higher Temperature

Hypothesis

Mustafar's temperature of 700 K is significantly higher than Earth's average temperature. This increases the local speed of sound and affects Mach number calculations.

Observation

Although the rocket achieved a higher velocity on Mustafar, the Mach number increased less dramatically because the speed of sound was also higher.

Effect of Wind

Hypothesis

A 10 m/s crosswind should increase weathercocking and downrange drift.

Observation

The simulated drift distance increased from approximately **185 m** under Earth conditions to approximately **340 m** under Mustafar conditions.

9. Recovery System Evaluation

Recovery Performance

Parameter	Value
Drogue Deployment Altitude	3999m
Main Chute Deployment Altitude	3999m
Payload Deployment Altitude	3999m
Final Descent Velocity	7.1m/s
Recovery Successful?	Yes

10. Conclusion

The two-stage sounding rocket was designed and simulated in OpenRocket using the specified Mustafar environmental conditions. The final design achieved an apogee of 3999 m while carrying a payload of 1.0kg. The vehicle maintained acceptable stability and completed recovery using parachute deployment.

THANK YOU